

EASTERN INTERNATIONAL UNIVERSITY**SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY**

Course Name: Introduction To Databases	Course Code: CSE 301
Practice Assignment - Quarter 3, 2025-2026	Lab 2

Student's Full Name: _____ Student ID: _____

ENTITY RELATIONSHIP MODEL TO RELATIONAL MODEL

Objectives

- Identify entities, attributes, keys, and relationships from business requirements
- Draw a conceptual er diagram correctly
- Convert an ER model into a Relational model
- Implement the Relational model in mysql

Guide

Stage 1. Conceptual ER Model: Identify entities, attributes, keys, relationships, and cardinalities from the business requirements. Do not use SQL data types, PK, FK, or table-style notation.

Stage 2. Relational Model: Convert the conceptual ER model into relations. For each relation, clearly specify the primary key and foreign key(s).

Stage 3. MySQL Physical Schema: Implement the relational model in MySQL using appropriate data types and constraints.

General Instructions

- Analyze the requirements carefully before drawing the diagram.
- Use singular names for entities, for example: Customer, Invoice, Employee.
- Use lowercase names with underscores for attributes and columns, for example: customer_id, invoice_date, unit_price.
- For each problem, complete all three deliverables: conceptual ER diagram, relational model, and MySQL schema.

Problem 1. Sales Management Database

A retail store needs a database system to manage customers, staff, products, invoices, and invoice details. The system stores information about customers, including customer ID, full name, address, phone number, date of birth, registration date, and total sales amount. A customer may or may not be a registered member. The system also stores information about staff, including staff ID, full name, hire date, and phone number. Each product has a product ID, product name, unit, country of origin, and selling price. When a purchase is made, the store issues an invoice. Each invoice records the invoice ID, invoice date, the staff member who created the invoice, and the customer who made the purchase if the customer is a registered member. A non-member customer can still make a purchase. Each invoice contains one or more invoice detail records, and each invoice detail records a product, quantity, and selling price at the time of sale.

Requirements

- Identify entities, attributes, and keys.
- Identify relationships and cardinalities.
- Draw the conceptual ER diagram.
- Convert the ER model into a relational model.
- Write MySQL DDL statements.

Notes

- The customer in an invoice may be optional because non-member customers are allowed.
- The attribute `total_sales_amount` may be considered a derived attribute.
- The selling price should be stored in `invoice_detail` to reflect the price at the transaction time.

Problem 2. Company Employee Management Database

A company needs a database system to manage employees, departments, projects, and employee dependents. Each employee has an employee ID, first name, last name, date of birth, gender, and address. Each employee works for exactly one department and is directly supervised by one manager, who is also an employee, although some employees may not have a supervisor. The company has many departments, and each department has a department ID, department name, and the date on which the department manager started managing the department. Each department is managed by one employee and may operate in one or more locations. The company also manages projects, and each project has a project ID, project name, and project location. Each project is controlled by one department. Employees may work on many projects, and each project may involve many employees. For each employee assigned to a project, the system records the number of working hours per week. The company also stores information about employee dependents, including dependent name, date of birth, gender, and relationship to the employee. In addition, the company distinguishes between full-time employees and part-time employees: full-time employees receive a monthly salary, while part-time employees receive an hourly rate.

Requirements

- Identify entities, attributes, and keys.
- Identify relationships and cardinalities.
- Draw the conceptual ER diagram.
- Convert the ER model into a relational model.
- Write MySQL DDL statements.

Notes

- `DepartmentLocation` should normally be modeled separately because one department can have multiple locations.
- The supervision relationship is a recursive relationship on `Employee`.
- The employee-project relationship is many-to-many and includes `working_hours_per_week`.
- `Dependent` may be modeled as a weak entity.
- Full-time and part-time employees must be modeled consistently.

Submission Requirements

Put all work for each problem into one Word file, then convert the Word file to PDF and submit only the PDF file.

- The submission must include: requirement analysis, conceptual ER diagram, relational model, and MySQL schema.
- All sections must be arranged in order and clearly labeled.
- Only one PDF file is accepted for submission.

Important Reminder

A diagram that already includes PK, FK, data types, NOT NULL, and table structure is not considered a conceptual ER diagram. **It belongs to the relational or physical design stage.**